$T : V \longrightarrow W$
$\mathcal{B}_V, \mathcal{B}_W$
$A = [T]_{\mathcal{B}_V, \mathcal{B}_W}$
$T^* : W^* \longrightarrow V^*$
$\mathcal{B}_W^*, \mathcal{B}_V^*$
$A^t = [T^*]_{\mathcal{B}_W^*, \mathcal{B}_V^*}$